

A Systematic Literature Review of MLOps Adoption in Software Development Companies: Attributes, Maturity, Benefits and Challenges

Esteban Arteaga
University of Cauca
GTI Research Group
Popayán, Cauca, Colombia
estebanben@unicauca.edu.co
<https://orcid.org/0000-0003-1634-066X>

César Pardo
University of Cauca
GTI Research Group
Popayán, Cauca, Colombia
cpardo@unicauca.edu.co
<https://orcid.org/0000-0002-6907-2905>

Elizabeth Suescún
EAFIT university
Grupo GIDITIC
Medellín, Antioquia, Colombia
esuescu1@eafit.edu.co
<https://orcid.org/0000-0001-7872-76382>

Vladimir Villarreal
Technological University of
Panama
Grupo GITCE - UTP
Chiriquí, Panamá
vladimir.villarreal@utp.ac.pa
<https://orcid.org/0000-0003-4678-5977>

Abstract— Software development companies (SDCs) have started to include Machine Learning (ML) within their projects, which has allowed ML models to not be left behind to the experimentation stage, but to move to the production stage, here Machine Learning Operations (MLOps) arises to decrease the gap between operations, development and data science teams. Therefore, this article presents a systematic literature review on the adoption of MLOps in SDCs, based on the analysis of 28 primary studies. The findings of this article reflect the efforts of the scientific community to define the principles, roles, artifacts, benefits, challenges, and crucial factors for the implementation of MLOps in SDCs.

Keywords—Machine Learning Operations (MLOps), Pipeline, Software Development, Machine Learning (ML), DevOps, Systematic Literature Review (SLR).

I. INTRODUCTION

The adoption of Machine Learning (ML) and Deep Learning (DL) is a growing trend in software development companies (SDCs). A key factor has been the remarkable progress in processing hardware, such as Graphics Processing Units (GPUs) and Tensor Processing Units (TPUs), which are capable of drastically decreasing the time required for training and processing. In addition, technologies such as Docker and Kubernetes have allowed ML models to move from the experimentation stage to the production stage, because, traditionally, bringing ML models to production was a complicated task since additional infrastructure was required. In this aspect, processing hardware such as GPUs and TPUs allow better and faster training of models, and containerization technologies such as Docker and Kubernetes allow faster deployments to production [1]. This technological confluence has generated that SDCs begin to migrate to an environment where ML coexists with traditional and/or agile development (traditional development being understood as development that does not incorporate ML within any of its processes); therefore, how products and/or services have been developed must evolve, since apart from code, ML projects depend on datasets, thus a definition of new roles, new activities, new artifacts that

traditional and/or agile methodologies do not cover is needed [2], [3].

Talking about ML in development necessarily implies talking about data science, which is defined as: “a discipline that combines statistics and programming focused on extracting values from data”, this statement is since to develop an ML project, data is needed, which is the cornerstone on which all ML or DL development processes are based. For this reason, various methodologies have been developed to meet the characteristics of data science projects, as mentioned by Ordoñez *et al.* [4], including CRISP-DM (CRoss-Industry Standard Process for Data Mining) and DDS (Data Driven Scrum), among others. However, these methodologies lack certain processes to effectively manage and complete data science projects, so companies may choose to resort to a more empirical approach. In [5], the author posits a critical challenge: 87% of data science projects fail to reach production, and 77% of companies encounter significant difficulties in adopting Big Data and artificial intelligence (AI) initiatives. The translation of data science project initiatives from the laboratory to the real world is a domain that necessitates prioritized consideration from both researchers and industry, as numerous challenges center on the development of ML algorithms. However, there is a paucity of focus on the deployment of ML models into production and the subsequent maintenance of these models.

As a result of the above, Machine Learning Operations (hereinafter MLOps) emerges as a paradigm and a set of practices that originate and extend from the principles and culture of DevOps. Among its similarities, we have: Continuous Integration (CI) and Continuous Deployment (CD) processes, automation for the development of rapid deployments, collaboration and communication between those involved and reduction of delivery time, however, ML projects not only depend on code, but also data, so MLOps differs from DevOps in terms of the nature of the final product, that is, while DevOps delivers software and produces deterministic results, MLOps delivers ML models and produces probabilistic outputs, this distinction has led to the introduction of two new terms:

continuous monitoring (CM) and continuous training (CT) [6], [7].

On the other hand, including an ML area in traditional and/or agile software development projects may require overcoming certain technical challenges, such as: (i) the incorporation of new processes and tools, (ii) the inclusion of professionals with the necessary soft and technical skills to support the execution of the processes, and (iii) the high cost associated with its implementation because specialized hardware units for ML are not cheap. Currently, although small and medium-sized enterprises (hereinafter SMEs) lead the industrial and commercial infrastructure in the vast majority of countries worldwide [8], these types of companies must mainly face challenges due to their cost, size, limited access to sources of financing, marketing difficulties, and low technological development due to the need to adopt quality models and standards, among others. [9]. In this sense, software SMEs have a limited capacity for the implementation of projects involving MLOps, mainly due to the high costs of implementation, deployment, and associated maintenance.

In addition, it is possible to find that software SMEs can also present chaotic processes, because most of them are defined in an empirical, incomplete, and ineffective way, since the development processes lack definition, control, and predictability. [8], [9] In this sense, and due to the nature and composition of software SMEs, the level of uncertainty in those that apply a MLOps approach in their projects, generates a high level of uncertainty, reason why it is necessary to clarify, define, restructure and include new process elements, among them: roles, soft competencies, technical competencies, tools, artifacts, activities, tasks, among others, that were generally being carried out in the traditional and/or agile software development process, and that now require the inclusion of an MLOps-oriented approach, but that according to the characteristics of software SMEs, is also applicable to this type of companies, which would allow them to produce and maintain high quality software, at low cost and in an effective way.

By the above, an analysis of the information was carried out using a systematic literature review (hereinafter SLR), where it has been possible to observe that efforts have been made in the definition of proposals and solutions related to the evaluation of MLOps. Considering the above, the rest of the paper is organized as follows: Section II presents the research protocol followed for the development of the SLR. Section III presents the execution of the information search. Section IV presents the different analyses of the results obtained in response to the research questions. Section V presents the discussion of the results, the main observations, the limitations of the review, and the significance of this research. Finally, Section VI presents the conclusions and future work.

II. RESEARCH PROTOCOL

For this study, we compiled the main research papers by conducting a Systematic Literature Review (SLR). This technique allows us to efficiently identify, evaluate, and analyze papers related to research problems. Therefore, an SLR facilitates the collection of primary research on a specific topic [10]. To carry out the design of this SLR, we followed the protocol proposed by Kitchenham and Charters [11] and Budgen

et al. [12], which provides guidelines for conducting a systematic review. The protocol involved the following stages: (i) definition of the research questions; (ii) conduct of the search for relevant papers; (iii) selection of primary papers applying inclusion and exclusion criteria; and (iv) extraction of information.

A. Research Questions

The main objective of this SLR was to establish the state of the art of a process that supports MLOps in small (11 to 50 employees) and medium (51 to 200 employees) SDCs, for which different research questions derived from the search objectives were posed. These questions were formulated based on the Goal-Question-Metric (GQM) approach described in [13]. This approach introduces a measurement model that is organized into three levels of abstraction: (i) conceptual level, where the objectives that will help to establish, understand and identify the purpose of the systematic review are defined; (ii) operational level, where questions derived from the objectives established at the conceptual level are developed, thus allowing to focus, characterize and structure the evaluation of the papers relevant to the topic of interest; and (iii) quantitative level, where specific metrics are created for each question, to provide quantifiable answers. Within this framework, a set of goals (measurement objectives) is developed to guide the conduct of the systematic review. The metrics associated with the GQM approach will be addressed in future work and are beyond the scope of this paper. The following are the search objectives (SO):

- **SO1:** Analyze the main scientific studies relating to the application of MLOps in small and medium software development companies, as evidenced in the current literature using the classification scheme proposed by Wieringa *et al.* [14], in addition to the definition of quality criteria for their classification according to their level of relevance.

- **SO2:** To provide support to academics and others interested in researching the application of MLOps in small and medium enterprises by gathering relevant information, including conceptual definitions, proposals, validations, processes, and research methods. This will allow us to identify the degree of progress of the initiatives based on the results obtained.

Based on the search objectives, three (3) research questions (hereinafter RQ) have been formulated, which are presented in Table I. In this table, the motivation behind each question and its relation to the established search objectives are detailed. These research questions are oriented to categorize the available information on MLOps in software development, as well as to identify the existing gaps in the research domain.

TABLE I. RESEARCH QUESTIONS

Id	Research question	Motivation	SO
RQ1	What are the essential attributes or principles for successful MLOps implementation?	Identify the essential attributes and principles of MLOps.	SO1
RQ2	How to measure the degree of maturity and/or capability of MLOps processes in a software company?	Determine how to classify the maturity level of a software development company applying MLOps.	SO2

RQ3	What are the benefits and challenges of adopting MLOps?	Determine the benefits and challenges involved in the adoption of MLOps in software development companies.	SO2
-----	---	--	-----

B. Search Strategy

The search was conducted between 2017 and 2025, because from this year the term MLOps became popular among the scientific community, the search was performed in the following databases: IEEE Xplore, Science Direct, Scopus, Google Scholar, Springer Link, and Web of Science. As a result, initiatives and papers related to the research topic were identified where the following were evidenced: conceptual definitions, terms, challenges, roles, artifacts, practices, activities, and other important approaches related to MLOps in the software development industry. On the other hand, for a first approach, other papers provided by experts classified as gray literature were considered. The search string defined for the review is presented in Table II. In addition, the following filters were applied: (i) papers published in English; (ii) papers exclusively from the following areas of knowledge: Computer Science, Software Engineering, and Software Development, and (iii) papers from journals or proceedings of scientific events.

TABLE II. SEARCH STRING

(“MLOps” OR “machine learning operation” OR “ML4DEV” OR “ML Production”) AND (“software Development”) AND (“agile” OR “agile release”) AND (“Adopt” OR “Integrate” OR “Integration” OR “Integrating”)

Adapted search strings for each database can be found in the following Zenodo link <https://zenodo.org/records/16906324>.

C. Selection of Primary Papers Applying Inclusion and Exclusion Criteria

Initially, the selection of relevant papers was based on a three-level review: (Level 1) review of the title, (Level 2) review of the abstract, introduction and conclusions, and (Level 3) review of the full text to determine whether the article met at least one of the inclusion criteria (IC) described in Table III. Subsequently, for the selection of primary papers, papers that met at least one of the exclusion criteria (EC) described in Table IV were discarded.

TABLE III. INCLUSION CRITERIA

Id	Inclusion criteria (IC)
IC1	Papers published in journals, congresses or conferences containing the words MLOps or its variants.
IC2	Papers published in journals, congresses or conferences on how to bring an ML model into production.
IC3	Papers published in journals, congresses or conferences that talk about how to maintain an ML model over time.
IC4	Studies that have been published in prestigious journals, congresses or conferences according to the quartile classification (Q1, Q2, Q3, Q4) with peer review.

TABLE IV. EXCLUSION CRITERIA

Id	Exclusion criteria (EC)
EC1	Duplicate papers.
EC2	Papers that do not address issues related to how to bring ML models into production.

EC3	Discussion-type papers or available only in the form of presentations or abstracts.
EC4	Papers that superficially address the research topic.
EC5	Papers that are books or book chapters.
EC6	Papers that are not written in English.
EC7	Papers that are not available in full text.

III. LITERATURE SEARCH PROCESS

To evaluate the relevance and quality of studies related to MLOps in small and medium-sized software development companies, a table of fifteen criteria was used, covering key aspects of both content and research impact. The criteria assess the study’s thematic focus, clarity in defining the research problem, coverage of the ML project lifecycle, and descriptions of required tools and technological artifacts. They also consider proposals for monitoring and continuous training of models in production, as well as methodological rigor, obtained results, and contributions to academic or industry. Additionally, the evaluation includes a discussion of research limitations, future works, and alternatives. The quality of the publication venue (based on journal or conference quartile ranking) and citation count (according to Google Scholar) are also considered. Each item is scored on a three-level scale (+1, 0, -1). The article extraction process is summarized in Fig. 1.

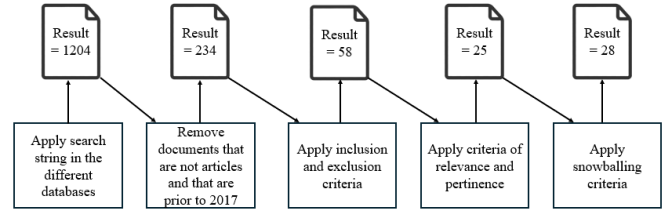


Fig. 1. Flow filtered SLR articles.

We can see that after applying the search string we found 1204 researches, of which only those articles published after 2017 were taken, then we applied the criteria of inclusion and exclusion for a total of 58 articles, then we applied criteria of relevance and relevance to further purify the information, reaching a total of 25 articles, finally after performing a snowballing process we reached a total of 28 articles. The list of primary articles can be viewed on Zenodo in the following repository <https://zenodo.org/records/15758020>.

IV. ANALYSIS OF RESULTS

A. Question 1: What are the essential attributes or principles for successful MLOps implementation?

One of the fundamental principles is *cross-disciplinary collaboration* (A1, A3, A25, A26), MLOps is not just the responsibility of data scientists; this principle requires the involvement of software engineers, operations experts, and other stakeholders. This collaboration ensures that machine learning models align with business objectives and are properly integrated into existing systems. Miscommunication between these groups can result in misunderstandings and the creation of solutions that do not meet business needs.

The adoption of *good programming practices* during model development is a critical attribute in ML environments (A16,

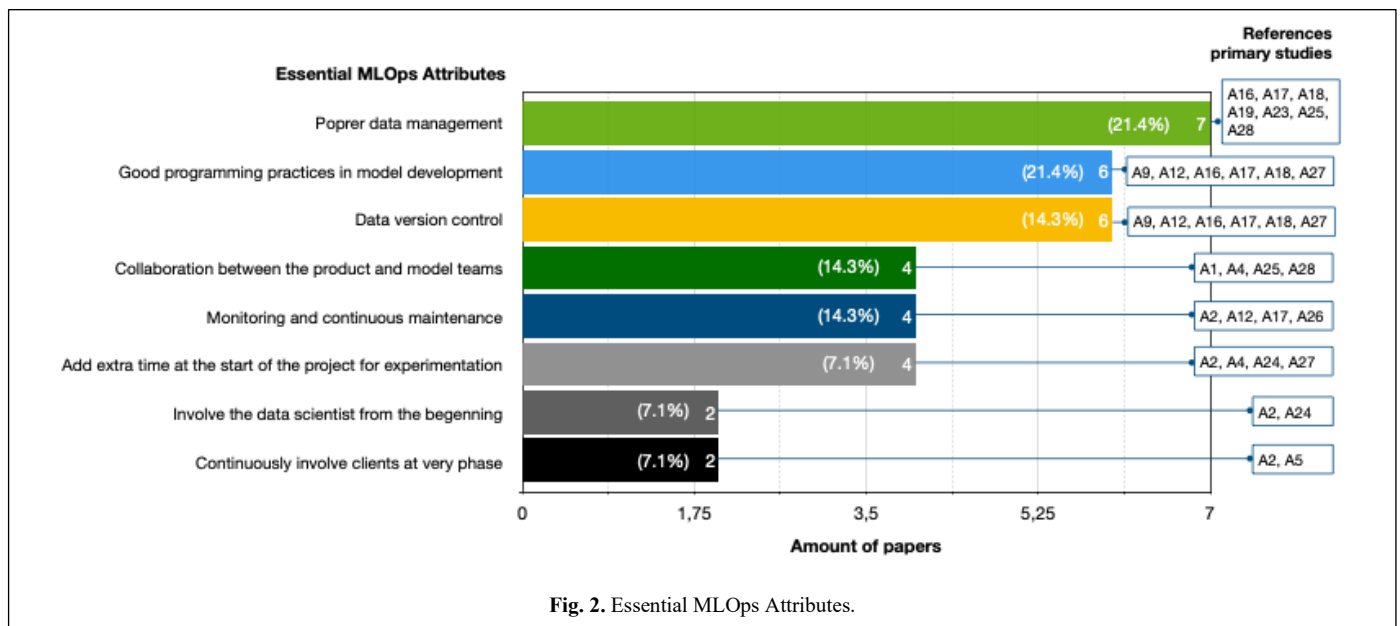


Fig. 2. Essential MLOps Attributes.

A17, A18, A19, A27). However, a significant proportion of data scientists lack formal training in computer science, leading to the omission of fundamental software development principles in the implementation of their models. This situation often results in a significant accumulation of technical debt. Consequently, although the developed models may be functional in the short term, their maintenance, scalability, and integration in production environments are compromised in the long term due to the lack of sustainable code structures. In this scenario, the intervention of ML engineers with software engineering training becomes essential, who, through refactoring processes and the application of robust programming standards, can guarantee the quality of the code and facilitate its deployment in production environments.

Moreover, *data management* is a critical component in the implementation of MLOps (A16, A17, A18, A19, A23), which is why the quality of the data used to train ML models directly impacts their performance. Therefore, it is essential to establish robust practices for data collection, cleaning, and storage. This includes the implementation of data pipelines that ensure that data is accessible and in a format suitable for use in ML models. Lack of proper data management can result in ineffective models and poorly informed business decisions.

Ongoing model *monitoring and maintenance* are critical components of successful MLOps processes (A1, A4, A25, A28). Once a model has been deployed in a production environment, it is crucial to systematically monitor its performance by tracking key performance metrics. This process makes it possible to detect deviations, model drift, or changes in data distribution (data drift), and to adjust or update promptly. To this end, it is necessary to define precise evaluation metrics that reflect the effectiveness of the model in the operational context, as well as to establish mechanisms that allow it to be updated in response to the evolution of data or to changes in the environment. The absence of a proactive monitoring strategy can lead to a progressive deterioration of the model's

performance, compromising the quality of predictions and the reliability of the system.

Organizational culture also plays a crucial role in the implementation of MLOps (A2, A3, A5, A24, A25), which is why fostering a culture that values experimentation, continuous learning, and adaptability can facilitate the adoption of MLOps practices. For this reason, continually involving the customer in each phase of MLOps and the data scientist can significantly reduce the time and expectations to deliver a final product.

Fig. 2 shows that most studies focus on the technical skills needed for success in MLOps, e.g., good programming practices, data version control, and proper data management, as well as the use of distributed technologies for data processing. However, very few studies consider soft skills, such as collaboration between the product and model teams and continuous customer and data scientist involvement. In this sense, data scientists must have solid technical skills as well as well-developed soft skills to explain technical processes to customers and different stakeholders in a way that they can understand. The list of primary articles can be viewed on Zenodo in the following repository: <https://zenodo.org/records/15758020>.

By analyzing the data found, it has been possible to show that in the time window consulted, as time progresses there are many more suggested attributes that are recommended to be taken into account to be successful when implementing MLOps, thus in the year 2022 the largest number of attributes were identified among which are: (i) adding additional time at the beginning of the project for experimentation, which may prove to be an analog to spikes within agile methodologies, also, (ii) it is suggested of vital importance to involve the data scientist from the beginning of the project, since it is problematic to follow a path that prioritizes the model without taking into account the product requirements, also, (iii) it is suggested to involve customers so that they can understand the progression of the project and be aware of possible missed deadlines.

B. Question 2: How to measure the degree of maturity and/or capability of MLOps processes in a software company?

To measure the degree of maturity and/or capability of MLOps processes in SDCs, it is essential to establish an assessment framework that considers various dimensions of the ML lifecycle. This framework should consider key aspects such as data management and governance, model quality and monitoring, process integration and automation (CI/CD/CT), as well as organizational culture and interdisciplinary collaboration. In this context, it is relevant to note that cloud service providers such as Azure and Google Cloud Platform (GCP) have developed maturity frameworks to help organizations assess and improve their MLOps capabilities. These frameworks offer structured guidelines to identify gaps in the implementation of MLOps practices and establish progressive improvement paths. The adoption of these models can be especially beneficial for companies seeking to systematize and optimize their ML-based solution development and implementation processes, enabling them to achieve higher levels of efficiency, quality, and sustainability in their operations. Table V presents a description of the maturity frameworks proposed in the software industry.

A clear relationship exists between the different frameworks for measuring MLOps maturity. While their levels vary, they all share the same goal: to achieve complete automation in maintaining Machine Learning (ML) models over time. The various benchmarks, such as those from Microsoft, Google, Kaisen, and GigaOm, structure the progression to a mature MLOps implementation in a comparable manner, showing a consensus on the direction of evolution. This progression starts with manual and ad-hoc processes, gradually introducing automation and DevOps/DataOps principles. Initially, the focus is on training, then deployment, until full lifecycle automation is achieved through CI/CD pipelines. On the other hand, there are

subtle differences in terminology and possibly in the granularity or emphasis of each stage. For example, Kaisen introduces the concept of “DataOps” and distinguishes between “Manual MLOps” and “Automated MLOps” before arriving at “Kaizen MLOps”. Google, conversely, strongly focuses on the notion of “Pipelines” (first ML and then CI/CD). Furthermore, Microsoft presents a clear progression from no MLOps to fully automated operations. GigaOm, with a strategy like Microsoft's, starts at level 0 with no DevOps practices and advances to level 4, where MLOps pipelines support frequent model updates (Continuous Training).

Understanding the different perspectives on MLOps maturity can help an organization assess its level and plan next steps. Adopting MLOps brings several significant benefits to organizations, along with challenges that need to be addressed to ensure successful implementation. We can observe that some maturity models start with automation that doesn't consider MLOps artifacts, while others begin by automating ML processes to achieve continuous training and deployment.

C. Question 3: What are the benefits and challenges of adopting MLOps?

The SLR identified both significant advantages and considerable obstacles associated with the adoption and implementation of MLOps practices in various organizations. To provide a structured view of these findings, Tables VI and VII summarize the key points drawn from the primary studies analyzed. Table VI details the main reported benefits, such as operational efficiency, cross-team collaboration, and the quality of deployed ML products. In contrast, Table VII outlines the most common challenges, ranging from technical issues such as data management and the experimental nature of ML to organizational challenges such as team integration, managing expectations, and a shortage of specialized talent.

TABLE V. MATURITY FRAMEWORKS PROPOSED BY THE SOFTWARE INDUSTRY

Company	Maturity levels
Microsoft	Microsoft proposes the maturity model based on 5 levels: - Level 0: No MLOps. At this stage, there are no MLOps practices in the company. Teams are disparate and releases are complicated. - Level 1: DevOps, but no MLOps: Here, the company may have adopted DevOps practices for traditional software development, but these practices have not yet been extended to ML workflows. - Level 2: Automated training. The training environment is fully managed; models are easy to replicate. - Level 3: Automated model deployment. At this stage, in addition to automated training, the deployment process of the trained models is also automated. This enables faster and more efficient deployment of the models into production. - Level 4: Full automated operations of MLOps. This is the highest level of maturity, where the operations of the ML models have been fully automated. This includes training, deployment, continuous monitoring and automated retraining of models in production (A7, A11, A25).
Google	Google proposes a maturity model based on 3 levels: - Level 0: Manual processes. At this initial level, MLOps processes are predominantly manual. - Level 1: ML pipeline automation: At this stage, automation of the ML pipeline is achieved. This involves automation of the steps involved in experimentation and model development. - Level 2: CI/CD pipeline automation: This is the most advanced level in the Google model, where automation of the Continuous Integration and Continuous Delivery (CI/CD) pipeline for ML models is achieved. This allows an automated and continuous implementation and update of models in production (A7, A11, A25).
Kaisen	According to A11, an empirical maturity model of MLOps based on 4 levels is proposed: - Level 0: Adhoc. This is the initial stage where data management processes are ad hoc, with improvised processes, non-standardized practices and documentation. - Level 1: DataOps: At this stage, standardized and automated processes for data management are implemented. - Level 2: Manual MLOps. At this level, companies manually standardize the end-to-end MLOps lifecycle. - Level 3: Automatic MLOps. At this stage, companies have full control of the end-to-end workflow in which they fully automate tasks related to data and models (A11).
GigaOm	According to A25, an MLOps maturity model based on 4 stages is proposed: - Level 0: No DevOps practices have been adopted. - Level 1: CI/CD DevOps practices are implemented for non-ML components. - Level 2: Iterative development, but no CI/CD for ML models yet. - Level 3: DevOps practices such as CI/CD are in place, plus code reviews for ML. - Level 4: Comprehensive MLOps pipelines supporting frequent model updates.

Furthermore, when comparing Table VI and Table VII, there is a clear disparity between the number of challenges and benefits identified in the adoption of MLOps: while the primary articles analyzed reveal a greater diversity of challenges (eight different categories), the benefits are grouped into only three main categories. This asymmetry does not necessarily imply that the challenges outweigh the benefits in terms of net impact, but it does suggest several hypotheses about the current state and perception of the MLOps field. First, this difference may be related to the maturity of the discipline, given that MLOps is still an emerging and rapidly evolving field, with an academic and practical body that is still in a consolidation phase, and where holistic conceptual frameworks that adequately integrate all its components and interactions are still lacking. Secondly, research bias could play a relevant role, since both academic and technical literature tends to focus on identifying problems,

formulating solutions, and delimiting critical areas that require further research and development.

Third, the intrinsic complexity of MLOps, integrating multiple disciplines such as data science, software engineering, operations, data management, ethics, and business, introduces a wide range of potential friction points that amplify the visibility of challenges. Finally, from a practical perspective, challenges often represent immediate and operational problems, e.g., faulty pipelines, data inconsistencies or lack of specialized personnel, making them recurring themes in reports and publications, while benefits tend to manifest themselves in a more strategic, incremental and less tangible way, thus being less reported in comparison.

TABLE VI. BENEFITS OF MLOPS ADOPTION IDENTIFIED IN THE LITERATURE

Benefits	Description	Reference
Faster implementation of Machine Learning models in production	One of the most significant benefits of MLOps is improved model lifecycle efficiency. By automating processes such as continuous integration and continuous deployment (CI/CD), organizations can reduce the time required to take a model from the development phase to production, enabling companies to respond more quickly to market demands and customer needs. MLOps, by applying DevOps principles to the ML lifecycle, enables a more agile and faster transition from experimental models to production environments.	(A1, A2, A3, A8, A9, A11, A14, A15, A23, A25, A26, A28)
Improved collaboration between data science and operations teams	Implementing MLOps promotes greater collaboration between teams from different disciplines, such as development, operations and business. This collaboration is essential to align technical objectives with business goals, resulting in more effective solutions tailored to customer needs. It fosters a working environment where knowledge and best practices are shared, which can lead to significant innovations.	(A3, A5, A9, A13, A14, A23, A26)
Improved Quality of models in production	MLOps encourages standardization of development processes, which contributes to the creation of more robust and reliable models. Implementing continuous monitoring and validation practices allows organizations to identify and correct errors more effectively, resulting in higher quality and more accurate models. In addition, the ability to continuously test and adjust ensures that models remain relevant and effective over time.	(A2, A4, A5, A23, A25, A26, A28)

TABLE VII. CHALLENGES OF MLOPS ADOPTION IDENTIFIED IN THE LITERATURE

Challenges	Description	Reference
Data quality and quantity	The creation of an MLOps model faces the fundamental challenge of data quality and quantity. DL techniques require large volumes of relevant data for training, and their robustness is often questioned. Data engineers and data scientists invest significant time and resources in cleaning, labeling and preparing the data, with low data quality being a common problem. Therefore, defining a clear data strategy is an essential practice within MLOps.	(A1, A5, A10, A22, A23, A24, A25, A27, A28)
Integrating data science and software engineering work	Data scientists often lack the software engineering background necessary for the maintenance and evolution of AI solutions. In addition, traditional software engineering patterns and practices do not always fit the unique needs of data science, such as data management and model drifting. Overcoming this gap requires multidisciplinary collaboration and adoption of MLOps to optimize the ML product lifecycle. Lack of shared understanding about the complexity of the product lifecycle can negatively impact the delivery of successful ML systems.	(A1, A2, A10, A12, A14, A16, A17, A18, A25, A27)
Unrealistic expectations about model capabilities	A lack of understanding of ML leads clients and managers to demand impossible results, such as a total absence of false positives or levels of accuracy that are unattainable with the available resources. In addition, there is a static view of model development, with no understanding of the need for continuous improvement and experimentation. Internally, data science teams often struggle to communicate the real limitations and potential of ML techniques to other project roles. This knowledge gap makes it difficult to define viable requirements and manage expectations throughout the project lifecycle.	(A2, A10, A18, A27)
Experimental nature of ML models	Unlike traditional software development, building ML systems involves greater uncertainty and risk due to the management of trained data and models. Practices such as managing experiments and testing multiple hypotheses are critical, reflecting an iterative and exploratory process to achieve desired goals. This experimental nature contributes to ML projects being more prone to failure, and organizations are often in the early stages of adopting practices to manage them effectively. Therefore, an adaptation of software engineering techniques and practices is required to address these particularities.	(A2, A3, A4, A5, A10, A11, A16, A18, A23, A25, A27)
Large amount of computational resources	The high cost of specialized hardware such as GPUs, and infrastructure hinders the implementation of ML especially for small and medium enterprises when considering the expected return on investment (ROI). In this aspect the cloud plays a key role by not requiring the purchase of expensive and specialized ML equipment, additionally this situation highlights the need for efficient resource management within MLOps practices and the exploration of less complex models in resource constrained environments.	(A3, A13, A17, A18, A19, A22, A23, A25, A27)
Protection of confidential data	It is critical to establish and manage customer privacy information from project conception. Organizations must monitor and observe model fairness, bias, and errors, considering the ethical and legal implications of data use. In addition, it is important to explain and give customers control over how their data is used and shared. Privacy and security management is not only a regulatory requirement, but also a key factor in building trust in ML systems.	(A3, A5, A16, A17, A24, A25, A27)
Shortages of qualified personnel	There is a lack of skilled individuals to fill the various positions demanded by MLOps, spanning crucial areas such as architecture, data engineering, ML and DevOps. This deficiency is directly related to inadequate training of future professionals, as data science curricula often do not incorporate MLOps. In addition, application domain expertise is required to extract value from data appropriately. The increasing importance of ML even for non-specialized roles further underscores this need for competent professionals.	(A3, A5, A10, A13, A17, A18, A22, A25, A27)
Long-term maintenance of the ML model	Updating models periodically by training them with recent data is a fundamental practice for mitigating obsolescence and counteracting performance degradation caused by data drift, i.e. changes in the distribution of data over time. In this context, it is critical to continuously monitor model behaviour in production environments for the early detection of performance deviations. Similarly, understanding and interpretability of the model play a key role in identifying and diagnosing operational anomalies, facilitating timely corrective action. Furthermore, the systematic evolution of the model, accompanied by recurrent evaluation processes, aligns with the best practices of an MLOps approach, thereby ensuring the model's robustness, reliability, and effectiveness throughout its operational lifecycle.	(A1, A5, A9, A13, A15, A16, A17, A18, A23, A25, A27, A28)

V. RESULTS AND DISCUSSION

This systematic review aimed to ascertain the current state of the literature on MLOps in SDCs, from which the following observations can be deduced:

- There is a high interest in the scientific community in researching the adoption of MLOps in SDCs, this due to the boom of data science, mainly at present where conventional and/or agile software development begins to converge with ML applications, therefore, it is necessary to deepen processes that allow us to quickly bring an ML model to production with the best features and the best possible quality.

- One of the biggest challenges of MLOps is the continuous monitoring and training, because ML models are susceptible to small changes such as data variation (drifting), which causes them to degrade easily and a new training is needed to meet the requirements, so this is a field of high interest that the scientific community is addressing because each retraining requires high computational resources which generates costs to organizations. [15], [16], [17].

- As mentioned in [15], [18], clear definition of roles in an MLOps process is critical to the successful implementation of ML models in production. Also, collaboration between different disciplines is essential, and each team member should have a well-defined role that contributes to the model lifecycle. This includes roles such as data scientists, data engineers, and DevOps, each with specific responsibilities that facilitate communication and efficiency. Similarly, in [17] it is mentioned that the specific roles of developers are not well defined and that developers often support the work to other people according to the needs of the project, which has led to a certain and overloaded convergence between the different roles of data scientists, ML engineers and software developers, for such reason it is necessary a clear definition of roles and responsibilities of each person involved in the MLOps process.

- Concerning the necessary infrastructure problem of MLOps, to achieve the process of integration and continuous deployment of an ML model to be taken to production, the use of tools like those used in DevOps has been proposed, such as the use of Jenkins and Kubernetes, specifically the use of Kubeflow [16], [19], which seeks to ease the burden of implementing MLOps workflows. Although Kubernetes can meet many of the requirements for implementing ML projects, it is considered a general-purpose framework that does not focus specifically on ML. In this sense it can be investigated in depth about the infrastructure needed to carry out an MLOps model.

- Generally, an MLOps process has several stages, which vary; however, to take an ML model to production, it is essential to follow several essential stages that ensure its effectiveness and sustainability. First, the problem definition is crucial, where the objectives are identified and business requirements are gathered, this process must be clear with the customer and the data scientist must be involved from the beginning, and realistic expectations must be explained to the customer [15], [20], [21], [22], [23]. Many of the studies mention particular cases that may affect SMEs, for example, the large amount of computational resources needed for training and deployment of models, which can be solved by using cloud resources (serverless) since the cost

of physical infrastructure for training ML models is very expensive [24], [25], [23]. Finally, the shortage of qualified personnel generates a drawback in software development companies and especially in small companies, since qualified personnel are required to create ML models that are sustainable over time (Continuous Monitoring/Continuous Training) [22], [23], and in many cases, data science programs do not include MLOps in their training [18], [25].

A. Limitations of the systematic literature review

The limitations of the search databases used prevented providing a more complete view of the state of the art on MLOps in software development companies. Additionally, the selection of primary documents may suffer from subjective bias, although measures were applied to mitigate it, such as data extraction through the cards. Furthermore, restricting the inclusion to articles in English may have excluded relevant publications in other languages. Nevertheless, this SLR has generated valuable results that may serve as a basis for future research.

VI. CONCLUSIONS AND FUTURE WORK

This SLR allowed the identification of relevant sources of information and research on the application of MLOps in software development companies. From the findings, it was possible to evidence the high interest of the scientific community, especially due to the popularity of AI in recent years.

MLOps can be defined as an instance of DevOps in which, through a collection of techniques and tools, the deployment of an ML model to production is achieved, applying the concepts of continuous integration and deployment present in DevOps, and applying new concepts such as continuous monitoring and training. Additionally, it identifies the main roles that must be taken into consideration, such as the data scientist, the ML engineer, the data engineer, the product owner, and the DevOps engineer.

Ongoing monitoring and training are among the most critical activities within MLOps projects because, once released to production, ML models begin to degrade due to the changing nature of the data. Therefore, great attention must be paid to monitoring model performance metrics and preventing models from becoming obsolete over time.

In general, several challenges have been identified in the implementation of MLOps in SDCs regardless of size (large, medium, small), such as the maintenance of ML models over time, unrealistic customer expectations, the problem of integrating data science and software development, data management, protection of confidential data, and the experimental nature of ML are some of the most common problems. When we talk about small and medium software development companies, the most representative problem corresponds to the cost of computing resources infrastructure and the shortage of qualified personnel in data science areas.

Many ML developments are carried out by data scientists, but most of them do not have computer science training, so they do not apply the best development and programming practices when creating ML models. This generates large amounts of technical debt in the form of code quality problems and the need

for rewriting in the future. In addition, data science programs do not include MLOps in their training. Similarly, ambiguity in defining roles and artefacts has led to many people involved in ML development being overloaded with activities, resulting in significant social debt.

As future work, it is proposed that the impact of technical debt caused when creating a productive ML model is identified, since many data scientists have no professional development in computer science or software engineering. Likewise, it is proposed that social debt is investigated, since many MLOps roles are not clearly defined, leading software engineers to perform activities outside their usual remit.

ACKNOWLEDGMENTS

Professor César Pardo thanks the University of Cauca for the institutional support provided as a full professor. Professor Vladimir Villarreal is a member of SNI in Panama.

REFERENCES

- [1] A. B. Kolltveit and J. Li, "Operationalizing Machine Learning Models - A Systematic Literature Review," *Proceedings - Workshop on Software Engineering for Responsible AI, SE4RAI 2022*, pp. 1–8, 2022, doi: 10.1145/3526073.3527584.
- [2] P. Ruf, M. Madan, C. Reich, and D. Ould-Abdeslam, "Demystifying MLOps and Presenting a Recipe for the Selection of Open-Source Tools," *Applied Sciences* 2021, Vol. 11, Page 8861, vol. 11, no. 19, p. 8861, Sep. 2021, doi: 10.3390/AP11198861.
- [3] D. A. Tamburri, "Sustainable MLOps: Trends and Challenges," *Proceedings - 2020 22nd International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNASC 2020*, pp. 17–23, Sep. 2020, doi: 10.1109/SYNASC51798.2020.00015.
- [4] A. Ordoñez, J. S. Rojas, J. Gomez, and G. Ramirez, "Methodology Based on MLOps (Machine Learning Operations) for Management Support in Data Science Projects," *Revista Colombiana De Tecnología Avanzada*, pp. 87–103, Sep. 2023, Accessed: Nov. 08, 2023. [Online]. Available: https://www.researchgate.net/publication/373654041_Methodology_Based_on_MLOps_Machine_Learning_Operations_for_Management_Support_in_Data_Science_Projects
- [5] I. Martinez, E. Viles, and I. G. Olaizola, "Data Science Methodologies: Current Challenges and Future Approaches," *Big Data Research*, vol. 24, May 2021, doi: 10.1016/J.BDR.2020.100183.
- [6] G. Symeonidis, E. Nerantzis, A. Kazakis, and G. A. Papakostas, "MLOps - Definitions, Tools and Challenges," *2022 IEEE 12th Annual Computing and Communication Workshop and Conference, CCWC 2022*, pp. 453–460, 2022, doi: 10.1109/CCWC54503.2022.9720902.
- [7] J. Klaise, A. Van Looveren, C. Cox, G. Vacanti, and A. Coca, "Monitoring and explainability of models in production," *Workshop on Challenges in Deploying and Monitoring Machine Learning Systems (ICML 2020)*, Jul. 2020, Accessed: Aug. 23, 2024. [Online]. Available: <https://arxiv.org/abs/2007.06299v1>
- [8] "Pequeñas y medianas empresas y creación de empleo decente y productivo," 2015. Accessed: Nov. 14, 2023. [Online]. Available: http://www.ilo.org/ilc/ILCSessions/previous-sessions/104/reports/reports-to-the-conference/WCMS_358292/lang-es/index.htm
- [9] M. Estayno, G. Dapozo, C. Greiner, L. C. Pletsch, and S. Pelozo, "Caracterización de las pymes de software de la región NEA orientada hacia un marco de mejora de la calidad," *XV Congreso Argentino de Ciencias de la Computación*, pp. 901–910, 2009.
- [10] K. Petersen, S. Vakkalanka, and L. Kuzniarz, "Guidelines for conducting systematic mapping studies in software engineering: An update," *Inf Softw Technol*, vol. 64, pp. 1–18, Aug. 2015, doi: 10.1016/J.INFSOF.2015.03.007.
- [11] B. Kitchenham and S. Charters, "Guidelines for performing Systematic Literature Reviews in Software Engineering," *EBSE Technical Report EBSE-2007-01 School of Computer Science and Mathematics, Keele University*, pp. 1–57, Jan. 2007.
- [12] D. Budgen, M. Turner, P. Brereton, and B. Kitchenham, "Using Mapping Studies in Software Engineering," *Annual Workshop of the Psychology of Programming Interest Group*, 2008.
- [13] V. R. Basili and G. Caldiera, "The Goal Question Metric Paradigm," *Encyclopedia of Software Engineering - 2 Volume Set*, vol. 2, pp. 528–532, 2000, Accessed: Aug. 24, 2024. [Online]. Available: <https://www.cs.umd.edu/~basili/publications/technical/T89.pdf>
- [14] R. Wieringa, N. Maiden, N. Mead, and C. Rolland, "Requirements engineering paper classification and evaluation criteria: A proposal and a discussion," *Requir Eng*, vol. 11, no. 1, pp. 102–107, Mar. 2006, doi: 10.1007/S00766-005-0021-6.
- [15] N. Nahar, S. Zhou, G. Lewis, and C. Kastner, "Collaboration Challenges in Building ML-Enabled Systems: Communication, Documentation, Engineering, and Process," *Proceedings - International Conference on Software Engineering*, vol. 2022-May, pp. 413–425, Oct. 2021, doi: 10.1145/3510003.3510209.
- [16] D. R. Niranjana and Mohana, "Jenkins Pipelines: A Novel Approach to Machine Learning Operations (MLOps)," *International Conference on Edge Computing and Applications, ICECAA 2022 - Proceedings*, pp. 1292–1297, 2022, doi: 10.1109/ICECAA55415.2022.9936252.
- [17] S. Laato, M. Mäntymäki, M. Minkinen, and D. Dennehy, "Machine Learning System Development in Information Systems Development Praxis," *Communications of the Association for Information Systems*, vol. 54, no. 1, pp. 133–154, Feb. 2024, doi: 10.17705/ICAIS.05406.
- [18] Y. Mehmood, N. Sabahat, and M. A. Ijaz, "MLOps critical success factors - A systematic literature review," *VFAST Transactions on Software Engineering*, vol. 12, no. 1, pp. 183–209, Mar. 2024, doi: 10.21015/VTSE.V12I1.1747.
- [19] J. Díaz-de-Arcaya, A. I. Torre-Bastida, R. Miñón, and A. Almeida, "Orfeon: An AIOps framework for the goal-driven operationalization of distributed analytical pipelines," *Future Generation Computer Systems*, vol. 140, pp. 18–35, Mar. 2023, doi: 10.1016/J.FUTURE.2022.10.008.
- [20] I. Martinez, E. Viles, and I. G. Olaizola, "Data Science Methodologies: Current Challenges and Future Approaches," *Big Data Research*, vol. 24, p. 100183, May 2021, doi: 10.1016/J.BDR.2020.100183.
- [21] N. Nahar, H. Zhang, G. Lewis, S. Zhou, and C. Kastner, "A Meta-Summary of Challenges in Building Products with ML Components - Collecting Experiences from 4758+ Practitioners," *Proceedings - 2023 IEEE/ACM 2nd International Conference on AI Engineering - Software Engineering for AI, CAIN 2023*, pp. 171–183, Mar. 2023, doi: 10.1109/CAIN58948.2023.00034.
- [22] AssresGebremariam, BhandariGuru, ShalaginovAndrii, GronliTor-Morten, and GhineaGheorghita, "State-of-the-Art and Challenges of Engineering ML- Enabled Software Systems in the Deep Learning Era," *ACM Comput Surv*, Nov. 2023, doi: 10.1145/3731597.
- [23] A. Hopkins and S. Booth, "Machine Learning Practices Outside Big Tech: How Resource Constraints Challenge Responsible Development," *AIES 2021 - Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 134–145, Oct. 2021, doi: 10.1145/3461702.3462527.
- [24] J. Diaz-De-Arcaya, A. I. Torre-Bastida, G. Zárate, R. Miñón, and A. Almeida, "A Joint Study of the Challenges, Opportunities, and Roadmap of MLOps and AIOps: A Systematic Survey," *ACM Comput Surv*, vol. 56, no. 4, p. 84, Oct. 2023, doi: 10.1145/3625289.
- [25] M. Zarour, H. Alzabut, and K. T. Al-Sarayreh, "MLOps best practices, challenges and maturity models: A systematic literature review," *Inf Softw Technol*, vol. 183, pp. 1–14, Jul. 2025, doi: 10.1016/J.INFSOF.2025.107733.